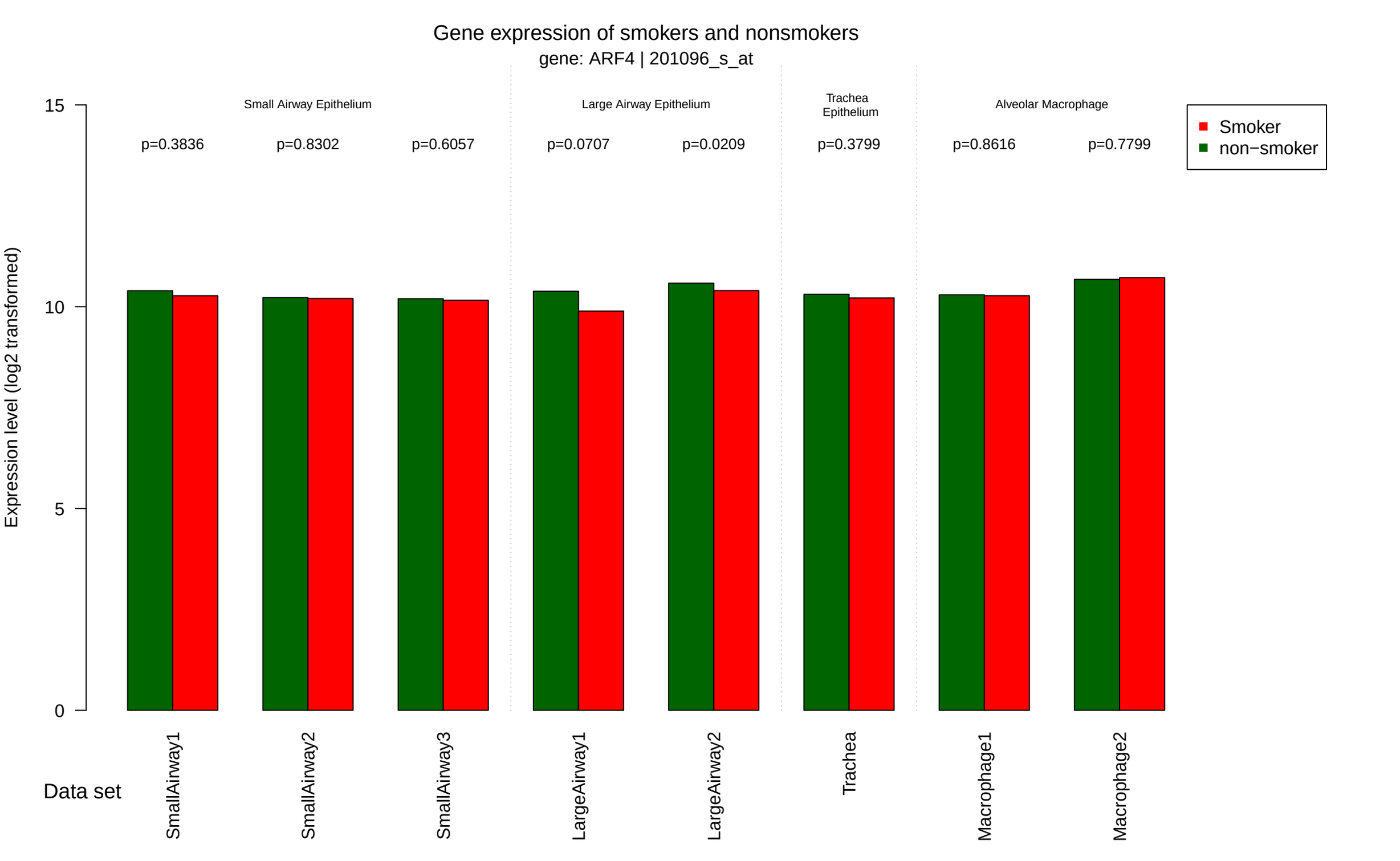
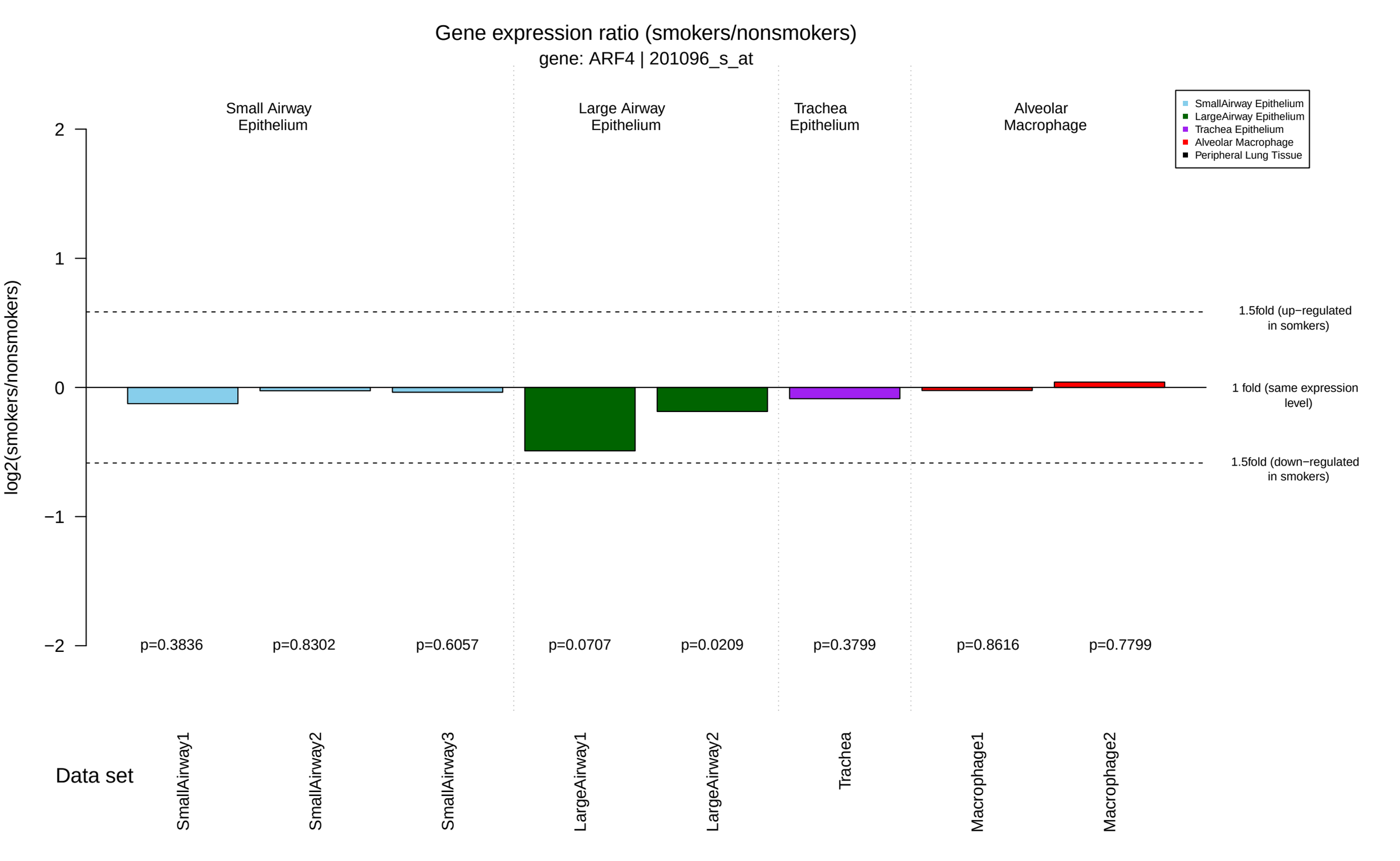
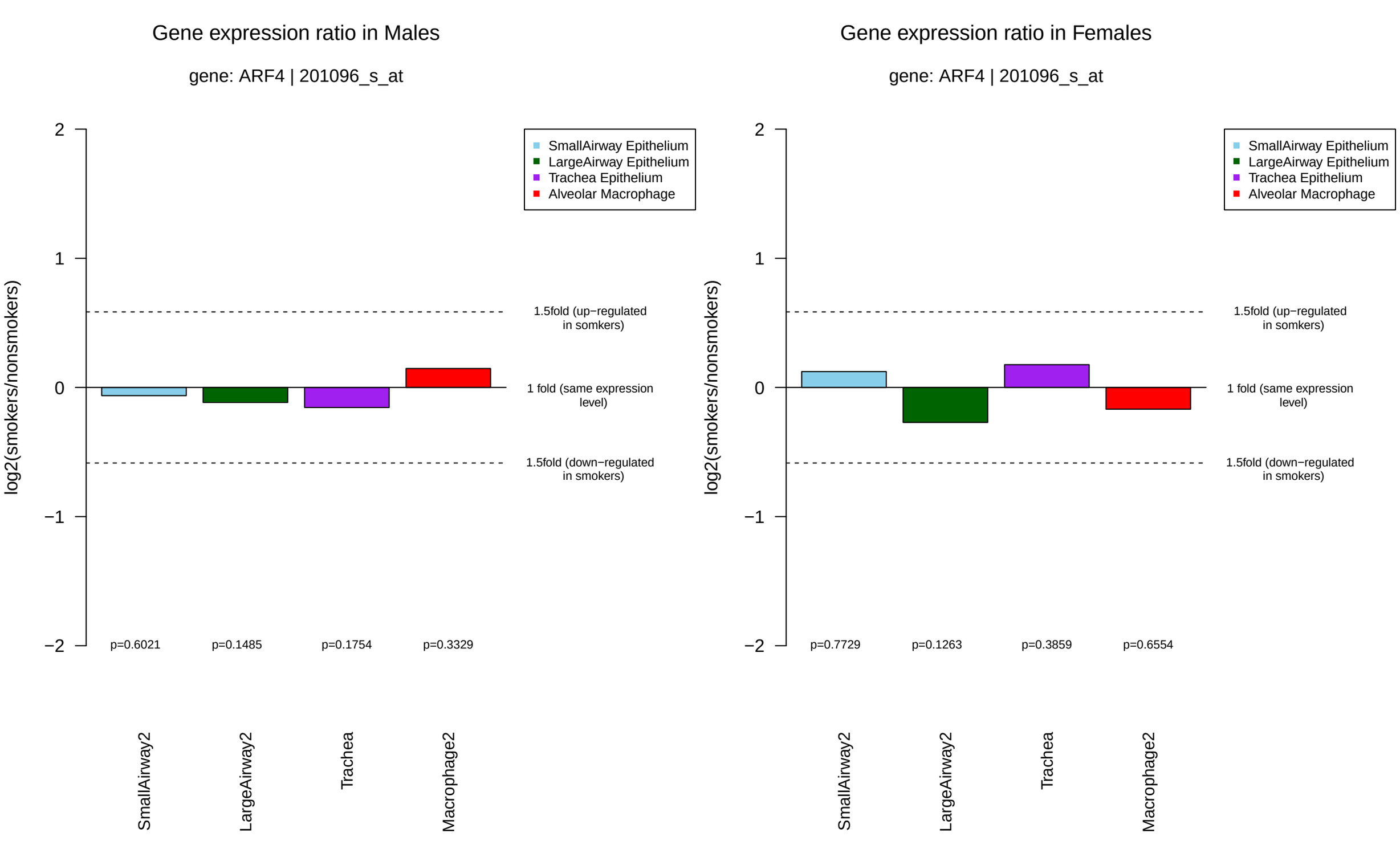
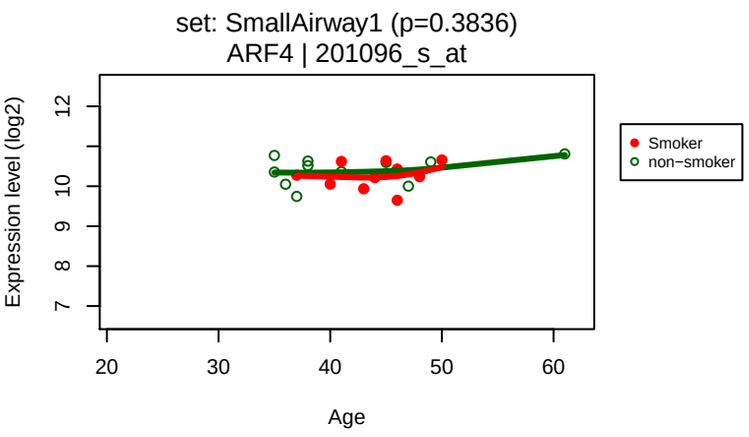
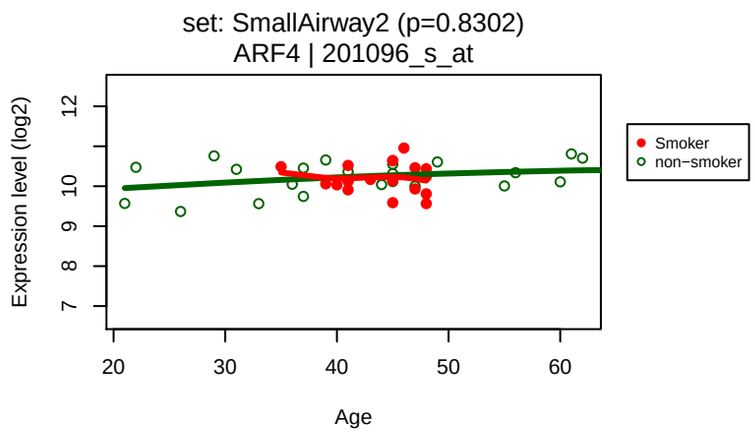
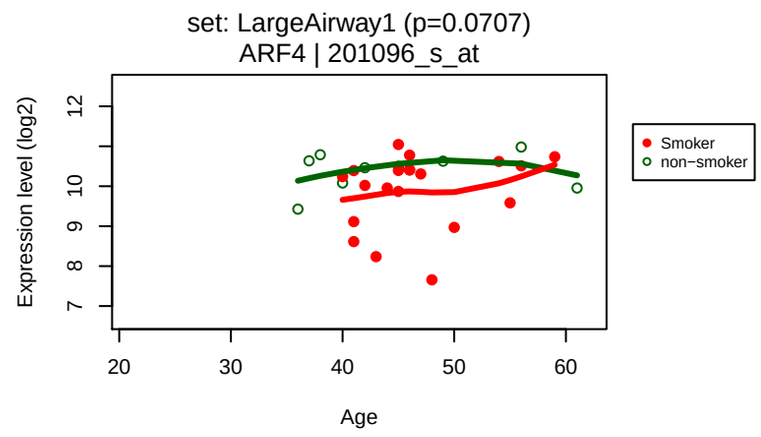

Gene expression of smokers and nonsmokers
gene: ARF4 | 201096_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.3836
p=0.8302
p=0.6057
p=0.0707
p=0.0209
p=0.3799
p=0.8616
p=0.7799
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

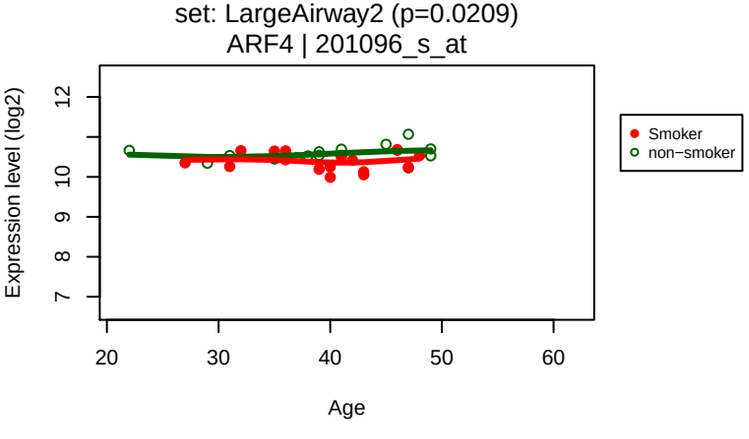

Gene expression ratio (smokers/nonsmokers)
gene: ARF4 | 201096_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.3836
p=0.8302
p=0.6057
p=0.0707
p=0.0209
p=0.3799
p=0.8616
p=0.7799
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

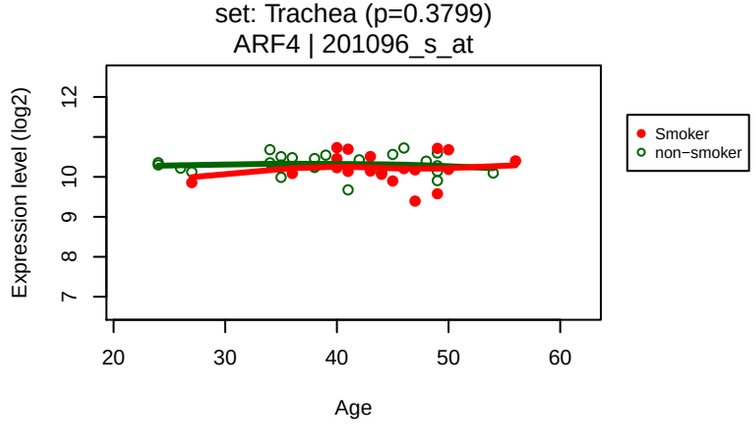

Gene expression ratio in Males
gene: ARF4 | 201096_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.6021
p=0.1485
p=0.1754
p=0.3329
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: ARF4 | 201096_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.7729
p=0.1263
p=0.3859
p=0.6554
SmallAirway2
LargeAirway2
Trachea
Macrophage2

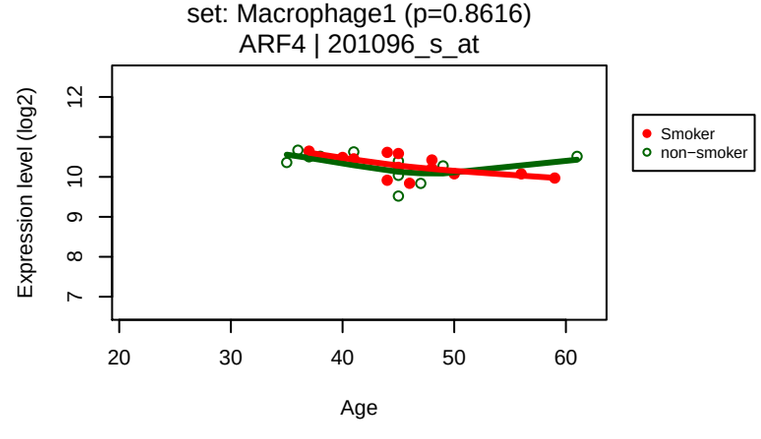

set: SmallAirway1 (p=0.3836)
ARF4 | 201096_s_at
Expression level (log2)
Age
Smoker
non−smoker

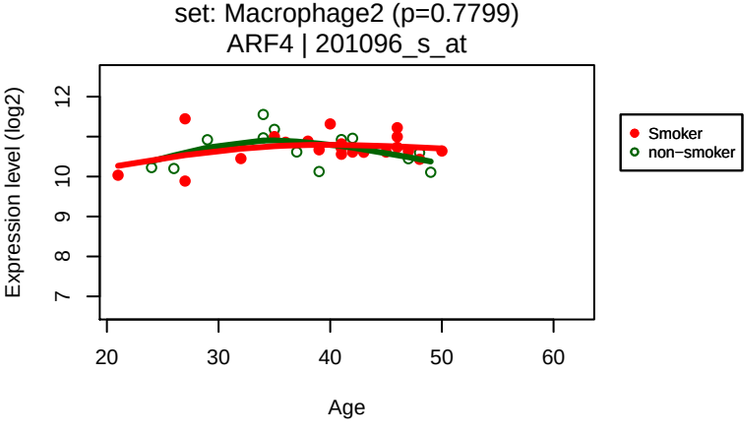

set: SmallAirway2 (p=0.8302)
ARF4 | 201096_s_at
Expression level (log2)
Age
Smoker
non−smoker

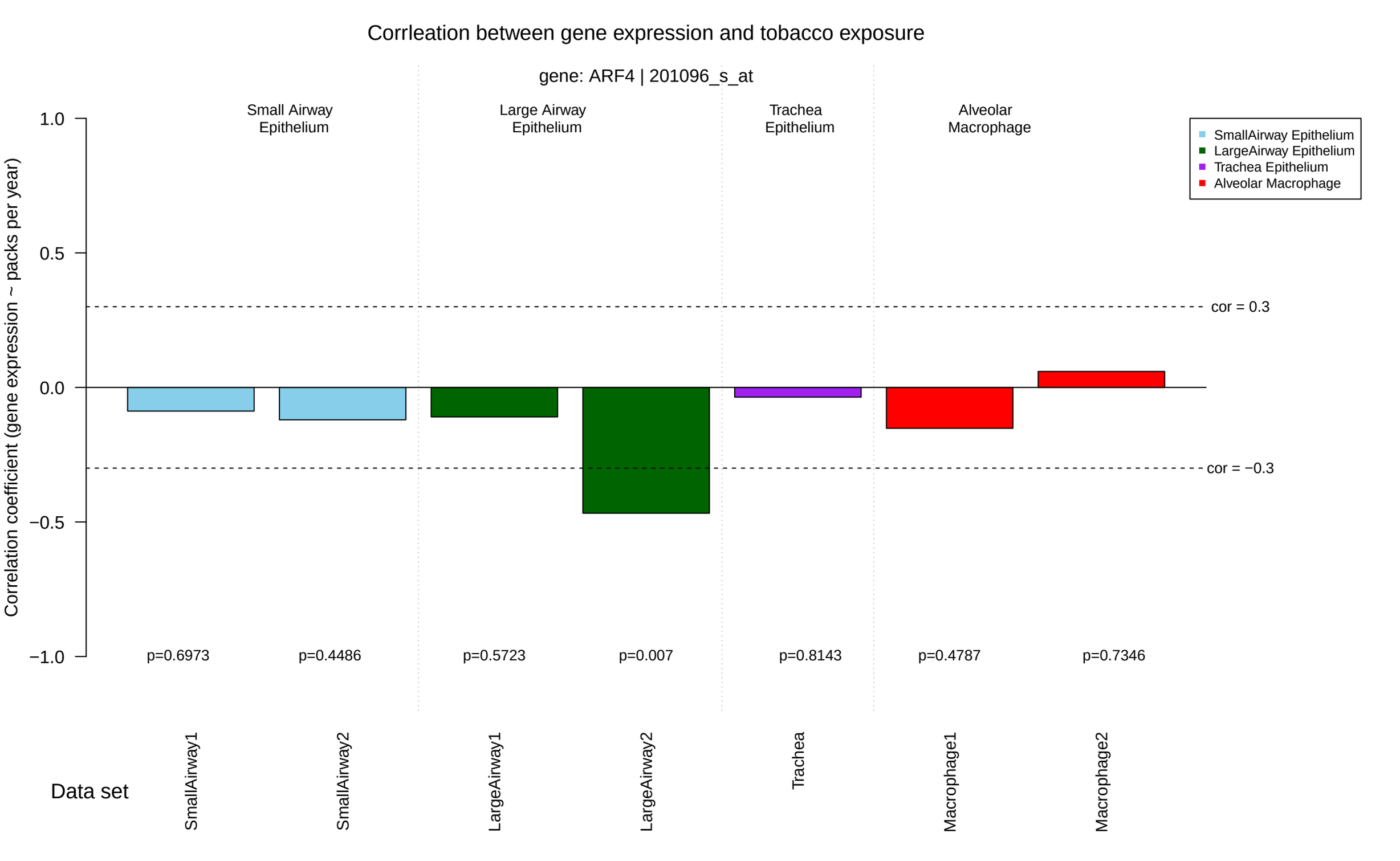

set: LargeAirway1 (p=0.0707)
ARF4 | 201096_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.0209)
ARF4 | 201096_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.3799)
ARF4 | 201096_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.8616)
ARF4 | 201096_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.7799)
ARF4 | 201096_s_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: ARF4 | 201096_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.6973
p=0.4486
p=0.5723
p=0.007
p=0.8143
p=0.4787
p=0.7346
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2